

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I Cheredla Jasvina/192324026 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

Answer :

Neat Sketch

Data Collection



Data Preprocessing



Feature Extraction



Model Training



Model Evaluation



Prediction

Functional Blocks

1. Data Collection:

Collect data from sources such as databases, sensors, websites, or cameras.

2. Data Preprocessing:

Clean the data by removing missing values, duplicates, and noise, and normalize it for better performance.

3. Feature Extraction:

Select important features that help the model learn effectively while removing unnecessary information.

4. Model Training:

Use a machine learning algorithm to learn patterns from the processed data.

5. Model Evaluation:

Test the trained model using metrics such as accuracy, precision, or recall to measure its performance.

Working with Examples

A. Supervised Learning (Labelled Data)

- Uses input data along with correct output labels.
- Example: **Email Spam Detection** (Spam / Not Spam).
- The model learns from labelled examples and predicts the class of new emails.

B. Unsupervised Learning (Unlabelled Data)

- Uses data without labels.
- Example: **Customer Segmentation** using K-Means clustering.
- The model automatically groups similar customers into clusters.

Labelled vs Unlabelled Data

- **Labelled Data:** Contains both input and output (e.g., image → Cat/Dog). Used in **supervised learning**.
- **Unlabelled Data:** Contains only input data. The model finds hidden patterns or groups. Used in **unsupervised learning**.

Conclusion:

A machine learning system processes data through **collection, preprocessing, feature extraction, training, and evaluation**. Supervised learning uses **labelled data** for prediction, while

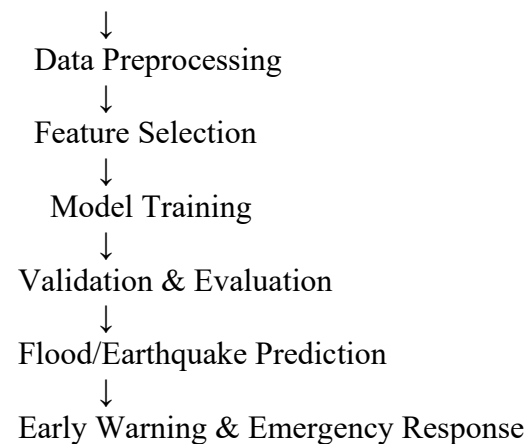
unsupervised learning uses **unlabelled data** to discover hidden patterns.

2. **Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response.**

Answer :

Neat Diagram

Historical & Real-time Data



1. Data Collection:

Collect historical records and real-time data such as rainfall, river water level, seismic activity, weather, and sensor data.

2. Data Preprocessing:

Remove missing values and noise, normalize the data, and combine historical and real-time information.

3. Feature Selection:

Choose important features like rainfall, river level, soil moisture, earthquake magnitude, and seismic readings to improve prediction accuracy.

4. Model Selection:

A **Neural Network (LSTM)** is suitable because it can learn patterns from time-series data and provide accurate predictions. For simpler datasets, algorithms like **Random Forest** can also be used.

5. Training, Validation, and Evaluation:

- Train the model using historical data.
- Validate it using unseen data to tune the model.
- Evaluate performance using metrics such as **Accuracy, Precision, Recall, and F1-score.**

Applications

- Provides **early warnings** before floods or earthquakes.
- Helps identify **high-risk areas.**
- Assists authorities in planning evacuations and resource allocation.
- Improves emergency response and reduces loss of life and property.

Conclusion:

A machine learning-based disaster management system combines historical and real-time data to accurately predict disasters. Using models like **LSTM Neural Networks** enables timely warnings, better risk assessment, and effective emergency management.

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

Answer :

Neat Diagram

Input Layer



Hidden Layer



Output Layer



Calculate Error



Backpropagation



Update Weights

Backpropagation is a learning algorithm used to train a multilayer perceptron (MLP). It updates the network weights by minimizing the prediction error.

Steps Involved

1. Forward Pass:

The input is passed through the hidden and output layers to produce a predicted output.

Example:

Input = 1, Actual Output = 1, Predicted Output = 0.8

2. Error Calculation:

The error is calculated as the difference between the actual and predicted outputs.

Error = Actual – Predicted = 1 – 0.8 = 0.2

3. Backward Propagation:

The error is propagated backward through the network, and the weights are adjusted using gradient descent to reduce the error.

Effect of Learning Rate

- A **small learning rate** gives slow but stable learning.
- A **large learning rate** speeds up learning but may overshoot the minimum and reduce accuracy.

Effect of Activation Functions

Activation functions introduce non-linearity, allowing the network to learn complex patterns.

Examples:

- **ReLU:** Faster training and commonly used in hidden layers.
- **Sigmoid:** Produces outputs between 0 and 1, suitable for binary classification.

Conclusion

Backpropagation improves the neural network by repeatedly performing **forward pass** → **error calculation** → **backward propagation** → **weight update** until the prediction error is minimized, resulting in better convergence and higher classification accuracy.

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

Answer :

Step 1: Selection

Choose the candidates with the highest fitness values.

Selected:

- Example 5 (Fitness = 80)
- Example 3 (Fitness = 75)

Step 2: Crossover

Combine the features of the selected parents to create a new offspring.

- Parent 1: Medium, High
- Parent 2: High, Medium

Offspring: High, High (Expected Fitness $\approx$ 90)

Step 3: Mutation

Slightly modify a feature to explore new solutions and avoid getting stuck in a local optimum.

Example:

- Speed remains High
- Accuracy improves to High (or another beneficial variation)

This increases the chance of finding a better solution.

Generation Improvement

Generation	Best Fitness
Initial	80
After Selection	80
After Crossover	90
After Mutation	92 (Improved)

Final Optimized Solution

- **Speed: High**
- **Accuracy: High**
- **Fitness Value: 92 (Approx.)**
- **Suitable Solution: Yes**

Conclusion

The selection process chooses the fittest individuals, crossover combines their good characteristics to produce better offspring, and mutation introduces small changes to explore new possibilities. After several generations, the Genetic Algorithm finds the optimal solution with High Speed, High Accuracy, and the highest fitness value.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

Answer :

Step 1: Training Using Backpropagation

A neural network takes Age, Blood Pressure, Glucose Level, and Lifestyle Risk as inputs.

During training:

- Forward Pass: Predicts whether the patient has the disease.
- Error Calculation: Compares predicted output with the actual output.
- Backpropagation: Updates the weights to reduce prediction error.
- This process repeats until the model achieves good accuracy.

Step 2: Confusion Matrix

From the given dataset:

Example Actual Predicted Result

1	Yes	Yes	TP
2	Yes	No	FN
3	No	No	TN
4	Yes	Yes	TP
5	No	Yes	FP

Therefore,

- True Positive (TP) = 2
- True Negative (TN) = 1
- False Positive (FP) = 1
- False Negative (FN) = 1

Step 3: Evaluation Metrics

- Accuracy = $(TP + TN) / \text{Total} = (2 + 1) / 5 = 60\%$
- Precision = $TP / (TP + FP) = 2 / 3 = 66.7\%$
- Recall = $TP / (TP + FN) = 2 / 3 = 66.7\%$

The model correctly predicts 3 out of 5 cases but makes 2 incorrect predictions.

Step 4: Role of Preprocessing and Feature Selection

- **Preprocessing:** Removes missing values, noise, and normalizes data, improving training quality.
- **Feature Selection:** Selects important features such as Blood Pressure, Glucose Level, and Lifestyle Risk, reducing unnecessary data and improving accuracy.

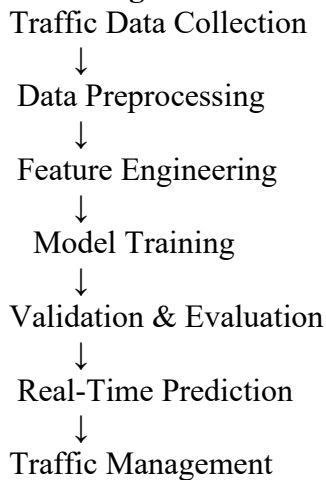
Conclusion

The neural network uses backpropagation to learn from errors and improve predictions. For the given dataset, TP = 2, TN = 1, FP = 1, and FN = 1, giving an accuracy of 60%. Proper preprocessing and feature selection make the model more accurate and reliable for disease prediction.

6. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making.

Answer :

Neat Diagram



1. Data Collection:

Collect data such as **vehicle density, time of day, weather conditions, and road type** from traffic cameras, sensors, and GPS devices.

2. Data Preprocessing:

- Remove missing and duplicate values.
- Normalize numerical data.
- Encode categorical data (weather, road type).
- Remove noise and outliers.

3. Feature Engineering:

Create useful features such as:

- Peak/Off-peak hours
- Average vehicle speed
- Traffic density
- Weather condition index

These features improve prediction accuracy.

4. Model Selection:

Random Forest is a suitable algorithm because it handles multiple features well, is accurate, and works effectively for traffic prediction. Neural Networks can also be used for very large datasets.

5. Training, Validation, and Deployment:

- **Training:** Train the model using historical traffic data.
- **Validation:** Test with unseen data and tune model parameters.
- **Evaluation:** Measure performance using **Accuracy, Precision, Recall, and F1-score**.
- **Deployment:** Deploy the trained model in a real-time traffic system to continuously predict congestion.

Applications

- Predicts traffic congestion in advance.
- Optimizes traffic signal timings.
- Suggests alternate routes to drivers.
- Helps authorities reduce traffic jams and improve emergency vehicle movement.

Conclusion

A smart traffic management system uses **vehicle density, weather, time, and road information** to predict congestion. With proper preprocessing, feature engineering, and a **Random Forest** model, the system provides accurate real-time predictions, enabling efficient traffic control and better road safety.

7. In the context of Convolutional Neural Networks (CNNs), explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance.

Answer :

Neat Diagram

Input Image



Convolution Layer



Pooling Layer



Fully Connected Layer



Output (Class Prediction)

1. Convolution Layer:

Applies filters (kernels) to the input image to extract features such as edges, corners, and textures. The output is called a feature map.

2. Pooling Layer:

Reduces the size of the feature maps while retaining important information. Max Pooling is commonly used to reduce computation and prevent overfitting.

3. Fully Connected Layer:

Combines the extracted features and performs the final classification of the image.

Feature Map Generation

A filter slides over the input image and performs convolution. The resulting values form a feature map, highlighting important image features.

Backpropagation in CNN

After prediction:

- Calculate the error between the predicted and actual output.
- Propagate the error backward through the network.
- Update filter weights using gradient descent.
- Repeat until the error is minimized.

Example

Image Classification:

Input: Image of a Cat

CNN extracts features like ears, eyes, and whiskers.

The fully connected layer classifies the image as "Cat."

Factors Affecting Performance

- Quality and size of the training dataset.
- Choice of activation function (e.g., ReLU).
- Learning rate.
- Number of convolution and pooling layers.
- Overfitting and regularization techniques.

Conclusion

A CNN consists of convolutional, pooling, and fully connected layers that automatically extract image features and classify images. Backpropagation updates the weights to improve accuracy, making CNNs highly effective for image recognition and classification tasks.

8. Consider the application of Genetic Algorithms for feature selection in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity.

Answer :

Neat Diagram

Initial Population



Fitness Evaluation



Selection



Crossover



Mutation



Best Feature Subset

1. Chromosome Representation:

In Genetic Algorithms, each chromosome represents a subset of features using binary values (1 = Selected, 0 = Not Selected).

Example:

Features: [Age, BP, Glucose, BMI]

Chromosome: [1, 0, 1, 1] → Selects Age, Glucose, and BMI.

2. Fitness Evaluation:

Each feature subset is evaluated by training a machine learning model. The subset with higher prediction accuracy and fewer features gets a higher fitness score.

3. Selection:

The best chromosomes with the highest fitness values are selected as parents for the next generation.

4. Crossover:

Selected parents exchange some of their features to create new offspring with better feature combinations.

5. Mutation:

Randomly changes one or more genes ($0 \leftrightarrow 1$) to introduce new feature combinations and avoid local optima.

Advantages :

Selects the most relevant features.

Reduces unnecessary data and computation.

Improves model accuracy.

Reduces overfitting and training time.

Conclusion :

Genetic Algorithms represent feature subsets as binary chromosomes and use selection, crossover, and mutation to evolve better feature combinations. This results in higher prediction accuracy, lower model complexity, and faster machine learning models.

9. In the context of classification problems, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection.

Answer :

Sample Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP = 30	FN = 10
Actual Negative	FP = 5	TN = 55

Calculation of Evaluation Metrics

- **Accuracy** = $(TP + TN) / \text{Total}$
= $(30 + 55) / 100 = 85\%$
- **Precision** = $TP / (TP + FP)$
= $30 / (30 + 5) = 85.7\%$
- **Recall (Sensitivity)** = $TP / (TP + FN)$
= $30 / (30 + 10) = 75\%$
- **F1-Score** = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
= 80% (Approx.)
- **ROC Curve:**
The ROC (Receiver Operating Characteristic) curve compares the True Positive Rate (Recall) with the False Positive Rate at different thresholds. A curve closer to the top-left corner indicates better classification performance.

Importance in Real-World Applications

- **Accuracy:** Indicates the overall percentage of correct predictions.
- **Precision:** Ensures that positive predictions are correct, making it useful in fraud detection, where false alarms should be minimized.
- **Recall:** Measures how many actual positive cases are identified, making it important in medical diagnosis, where missing a disease can be dangerous.
- **F1-Score:** Balances both precision and recall, especially useful when the dataset is imbalanced.
- **ROC Curve:** Helps compare different classifiers and select the best model.

Conclusion

Classification metrics provide different views of model performance. Accuracy measures overall correctness, Precision focuses on prediction quality, Recall measures the ability to detect positive cases, F1-score balances both, and the ROC curve evaluates the model across different thresholds. Together, these metrics help build reliable machine learning systems for applications such as disease detection and fraud detection.

10. Consider the use of Genetic Algorithms for optimization problems such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently.

Answer :

Neat Diagram

Initial Population



Chromosome Encoding



Fitness Evaluation



Selection



Crossover



Mutation



Optimal Schedule

Chromosome Encoding:

Each possible task schedule is represented as a chromosome, where each gene represents a task.

Example:

Tasks: T1, T2, T3, T4

Chromosome: [T2, T1, T4, T3]

2. Fitness Evaluation:

Each schedule is evaluated based on factors such as minimum completion time, resource utilization, and task priority. Higher fitness indicates a better schedule.

3. Selection:

The schedules with the highest fitness values are selected as parents for the next generation.

4. Crossover:

Two parent schedules exchange some tasks to create new offspring with better scheduling.

Example:

Parent 1: [T2, T1, T4, T3]

Parent 2: [T1, T3, T2, T4]

Offspring: [T2, T3, T4, T1]

5. Mutation:

Randomly changes the position of one or more tasks to generate new schedules and avoid local optima.

Advantages

- Finds an optimal task schedule.
- Reduces execution time.
- Improves resource utilization.
- Handles complex scheduling problems efficiently.

Conclusion

Genetic Algorithms encode task schedules as chromosomes and improve them through selection, crossover, and mutation. After several generations, the algorithm produces an optimized schedule with better performance and efficient resource utilization.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand